

WASP INTEGRITY

INSPECTION FINDS. INTEGRITY DECIDES.



WASP's Integrity service provides Level 1, 2 and 3 Fitness for Service assessments to API 579-1 / ASME FFS-1 — the globally recognised standard for determining whether ageing or damaged pressure equipment, piping and pipelines are fit for continued operation, and on what terms. The result is a clear, defensible answer: run, repair, or replace — backed by full engineering justification and documented to a standard that satisfies regulators, insurers and internal governance alike



THE NATURAL NEXT STEP AFTER INTELLIGENT PIGGING

The WASP UltraSight™ intelligent pigging inspection provides the high-density inspection data that accurate Fitness for Service relies on. With every defect precisely sized and positioned, API 579 assessments are based on measured reality—not assumptions—resulting in precise, less conservative engineering decisions.



ALL THREE LEVELS OF ASSESSMENT

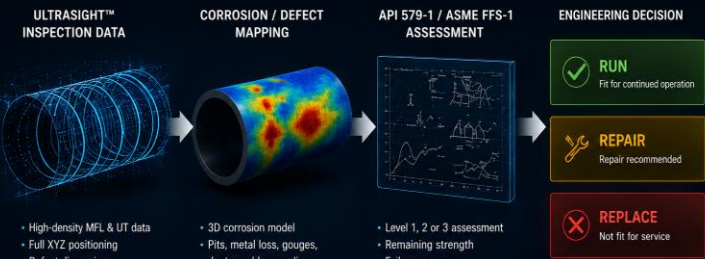
We perform Level 1, 2 and 3 Fitness for Service assessments to API 579-1 / ASME FFS-1, selecting the appropriate level of analysis for each asset. From rapid conservative screening to detailed stress analysis and advanced finite element modelling, every assessment is tailored to the complexity of the damage and the criticality of the equipment.



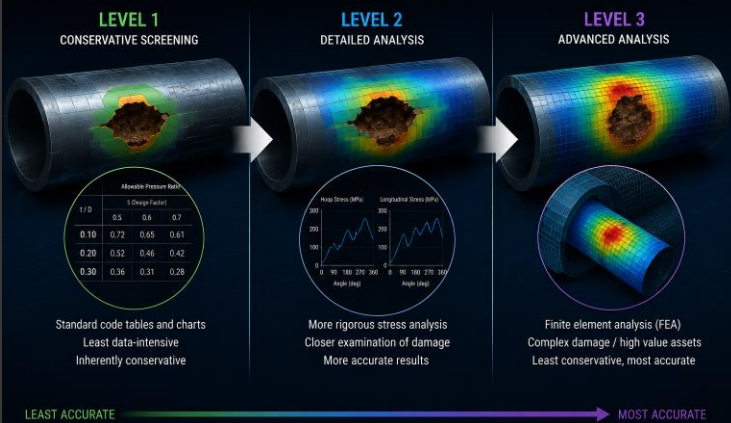
STANDALONE SERVICES

Fitness for Service starts with inspection data—not necessarily ours. We routinely perform API 579 assessments using data from third-party intelligent pigging, handheld UT surveys and historical inspection records, enabling operators to make full use of existing inspection information. — wherever the data came from, we can work with it.

ULTRASIGHT DATA TO INTEGRITY DECISION WORKFLOW



LEVEL 1, 2 AND 3 ASSESSMENT PROGRESSION



STANDARDS & CODES WE WORK WITHIN

■ API 579-1 / ASME FFS-1

The primary Fitness for Service standard governing all Level 1, 2 and 3 assessments for pressure equipment, piping and pipelines.

■ ASME B31.3

Process piping design code. All FFS assessments on process piping are framed within B31.3 allowables and acceptance criteria.

■ API 570

Piping inspection code. Defines the conditions under which Fitness for Service assessment is required and what it must demonstrate.

■ API 530

Remaining life calculation for fired heater and furnace process tubes, applied directly to UltraSight furnace inspection data.

■ ASME B31G

Widely used method for determining the remaining strength of corroded pipelines. Applied as an initial corrosion assessment before escalating to full API 579 Level 2 or 3 analysis where required.

■ API 1163

Inline inspection systems qualification standard. Underpins the integrity of the inspection data feeding into every assessment.

RUN. REPAIR, REPLACE?

BACKED BY FULL ENGINEERING JUSTIFICATION.



DAMAGE MECHANISMS WE ASSESS

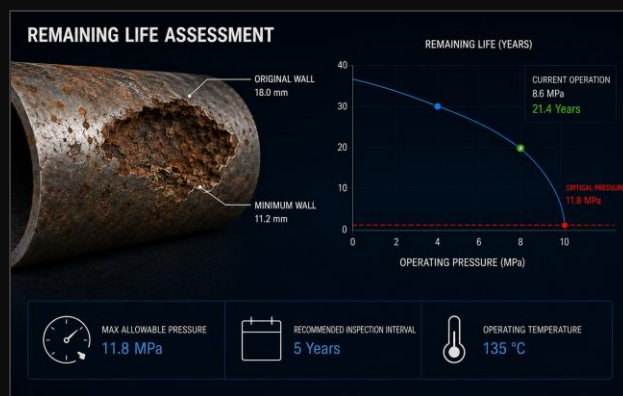


WHY OR WHEN IS FITNESS FOR SERVICE NEEDED?

- Future maintenance outage planning
- Defects identified during inspection
- Plant life extension or change of service
- Asset recommissioning
- Regulatory or insurer-driven assessment requirements
- Post-fire or post-incident evaluation
- Equipment operating beyond original design life

REMAINING LIFE ASSESSMENT

Remaining life assessment is the quantitative output that gives Fitness for Service its practical value — converting a pass/fail determination into a forward-looking engineering decision. It transforms inspection data into a projected operational timeline, giving operators a defensible basis for continued operation, a planned inspection interval, and an early warning of when the next decision point is approaching. For ageing assets, it is the difference between reactive crisis management and planned, risk-informed integrity management.



WHAT'S IN THE INTEGRITY REPORT?

- ✓ Identification and classification of all damage mechanism(s) present
- ✓ Full cross-referencing to intelligent pigging inspection report (where applicable)
- ✓ Calculated remaining strength and remaining life
- ✓ Recommended future operating conditions, including any pressure or temperature derating
- ✓ Full Fitness for Service Assessment results at corresponding levels
- ✓ Recommended inspection intervals and monitoring methods

EXPERIENCE YOU CAN TRUST

Our assessments are led by Chartered Engineers and PhD-qualified specialists in structural integrity, materials science and metallurgy — expertise few organisations can match under one roof. A Chartered Engineer-led report carries professional accountability that regulators, insurers and auditors recognise, making every conclusion legally and technically defensible.

What sets us apart is combining integrity expertise with deep practical knowledge of intelligent pigging. Our engineers understand not just what the data shows, but how it was acquired and how to interrogate it correctly — ensuring every dimension fed into an API 579 calculation reflects the true condition of the asset. Our metallurgists understand the degradation mechanisms driving each defect. Measuring a defect and understanding it are not the same thing — and the assessment, the monitoring plan and the remaining life calculation all depend on getting both right.

With over 75 years of combined experience across 40+ countries, you can trust WASP to give you the certainty your assets demand.